AUTONOMOUS NAVIGATION AND PATH PLANNING

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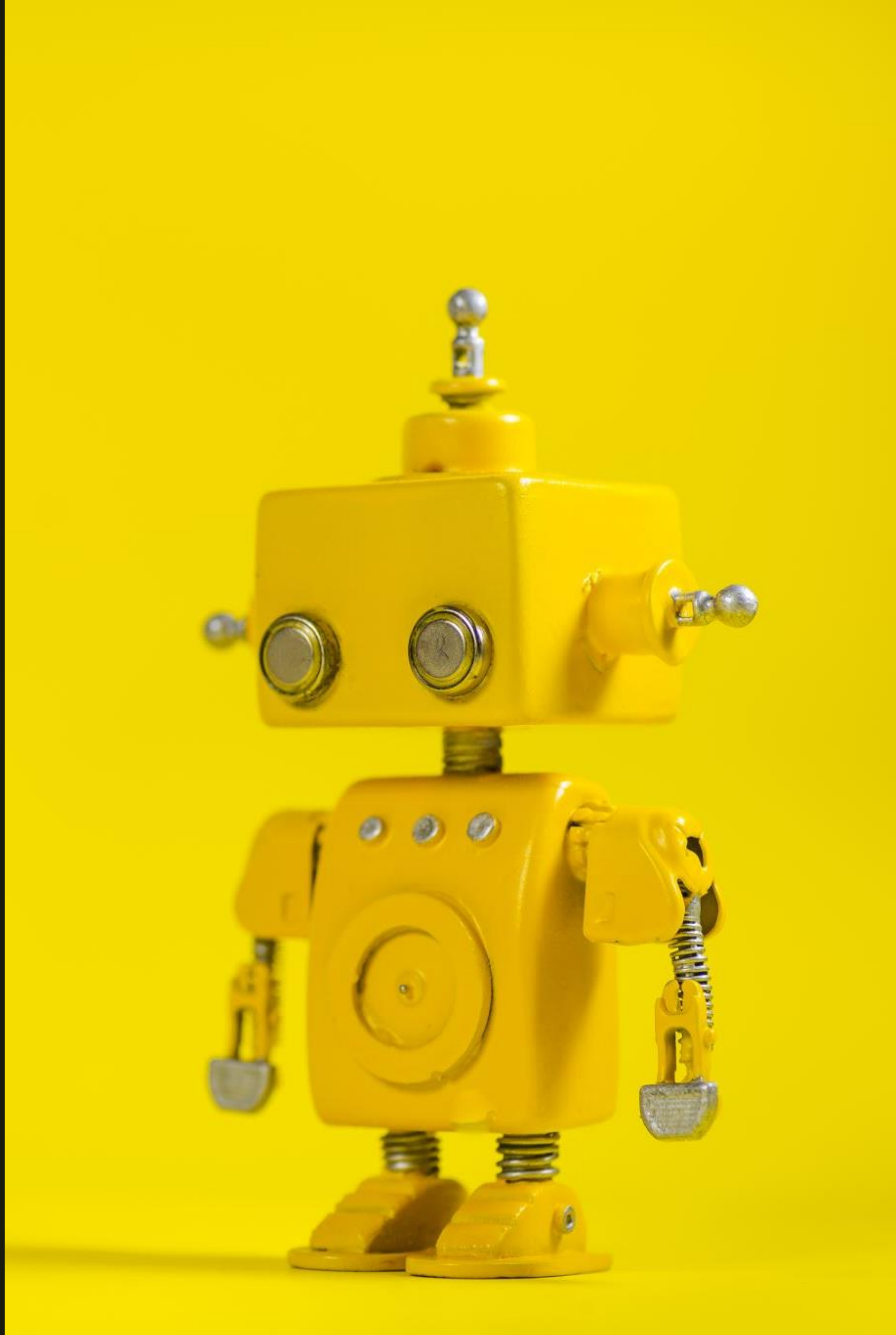
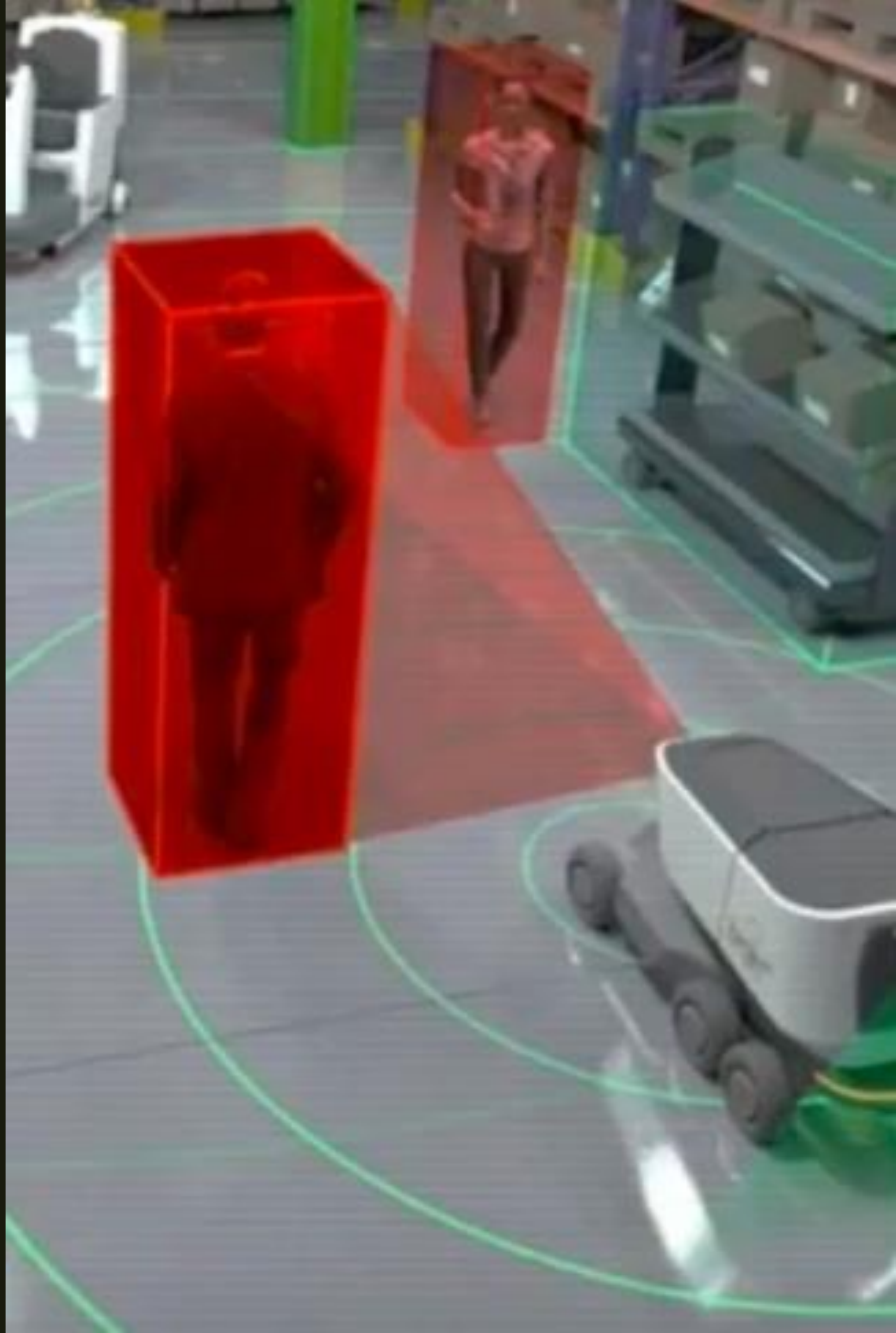
Agenda

- Introduction to Autonomous Navigation
- Key Components of Autonomous Navigation
- Localization Techniques
- Mapping Techniques
- Path Planning Algorithms
- Challenges in Autonomous Navigation
- Summary
- Q&A



Introduction to Autonomous Navigation

- Autonomous navigation refers to the ability of a robot to plan and execute a path from a starting point to a destination without human intervention.
- Essential for applications in robotics such as autonomous vehicles, drones, and service robots.

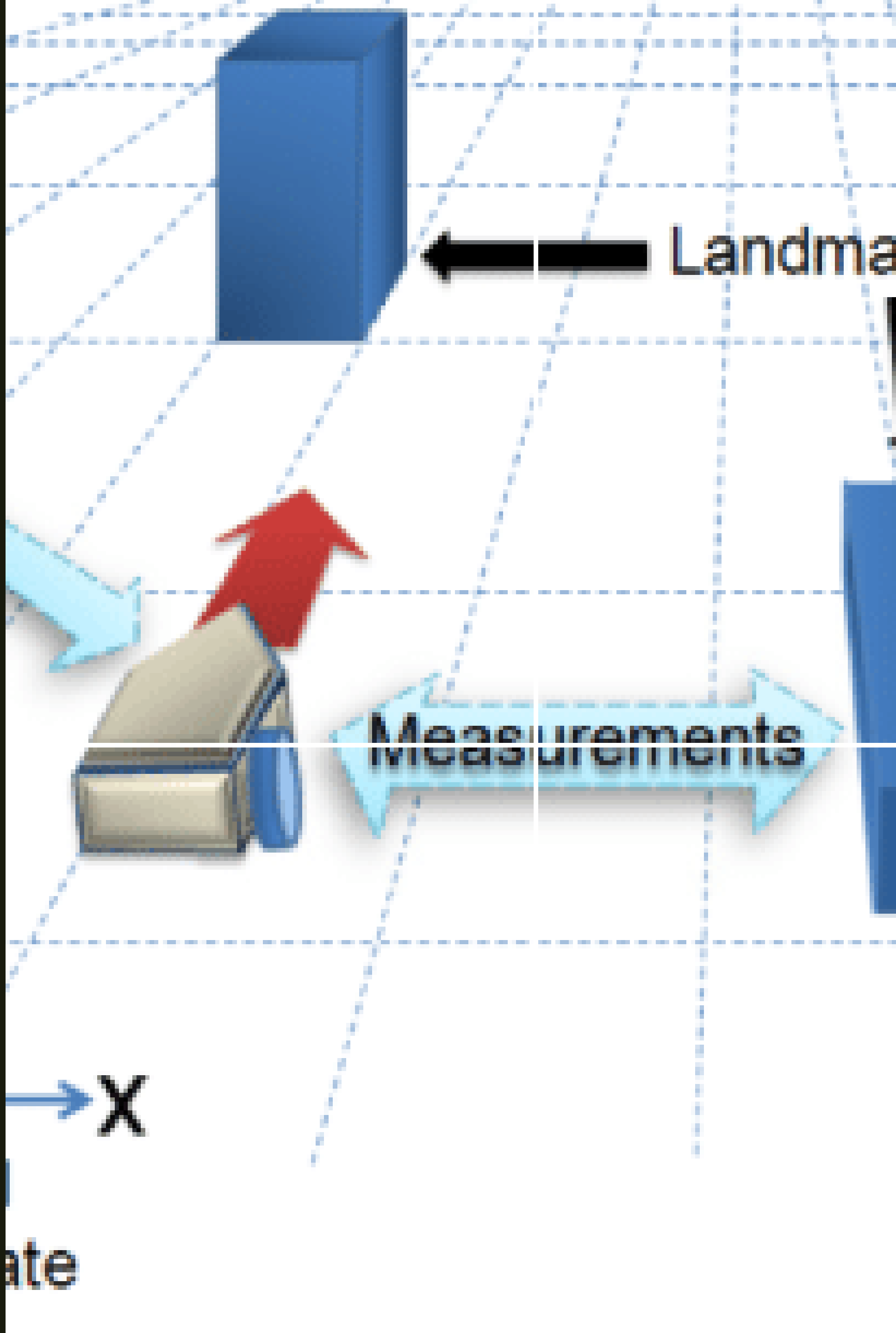


Key Components of Autonomous Navigation

- **Perception:** Sensing and interpreting the environment.
  - ***Sensors:*** *Cameras, LiDAR, radar, ultrasonic sensors.*
- **Localization:** Determining the robot's position within the environment.
  - ***Techniques:*** *GPS, SLAM (Simultaneous Localization and Mapping).*
- **Mapping:** Creating a representation of the environment.
  - ***Types:*** *Grid maps, topological maps, metric maps.*
- **Path Planning:** Finding a safe and efficient route to the destination.
  - ***Algorithms:*** *A*, Dijkstra's, RRT (Rapidly-exploring Random Trees).*
- **Motion Control:** Executing the planned path.
  - ***Controllers:*** *PID controllers, MPC (Model Predictive Control).*

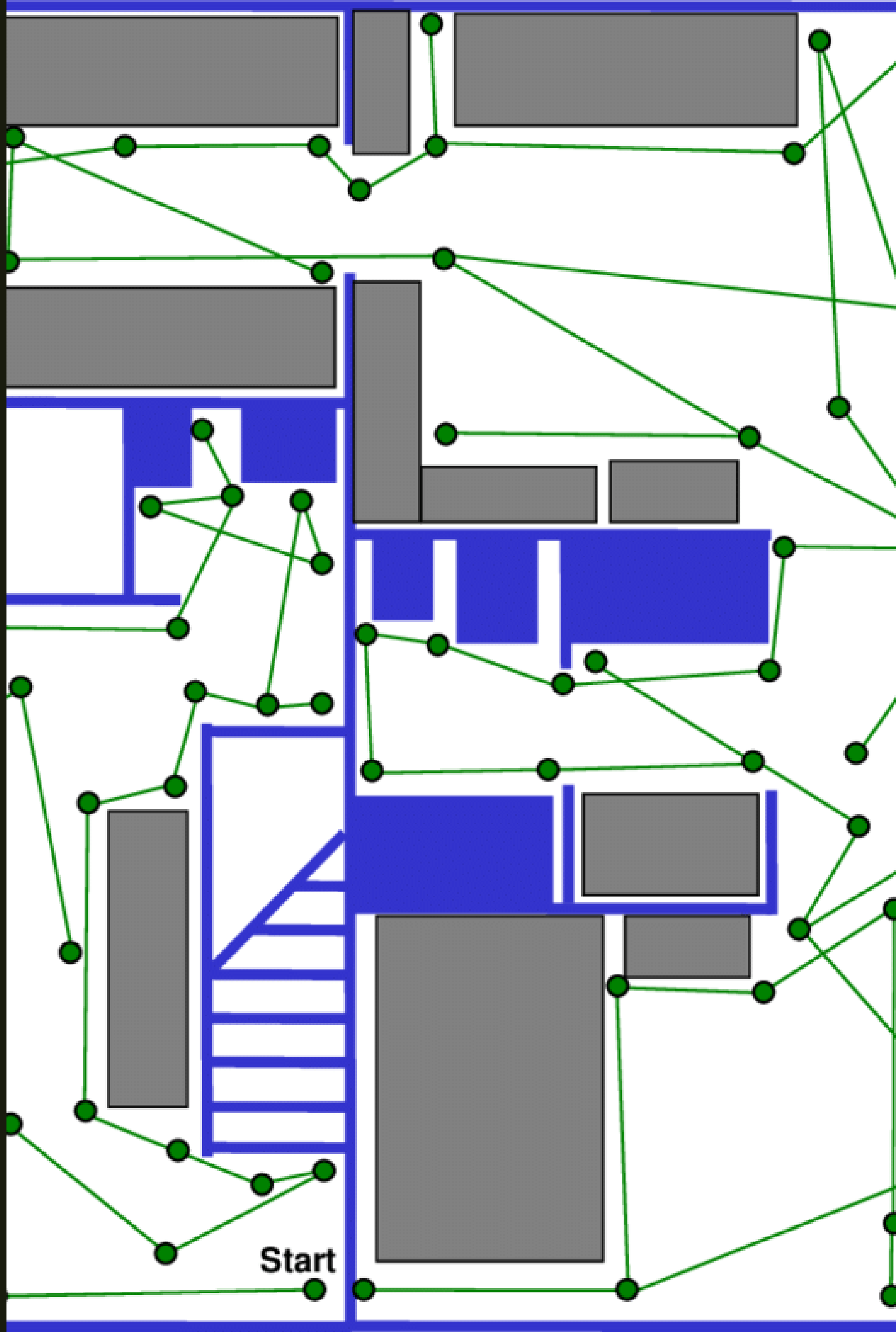
Localization Techniques

- GPS (Global Positioning System):
  - *Provides global location data using satellite signals.*
  - *Limited accuracy, signal loss in indoor environments.*
- Odometry:
  - *Estimates position based on wheel rotations or leg movements.*
  - *Accumulates errors over time (drift).*
- SLAM (Simultaneous Localization and Mapping):
  - *Builds a map of the environment while simultaneously tracking the robot's location within it.*
  - *Visual SLAM, LiDAR-based SLAM.*



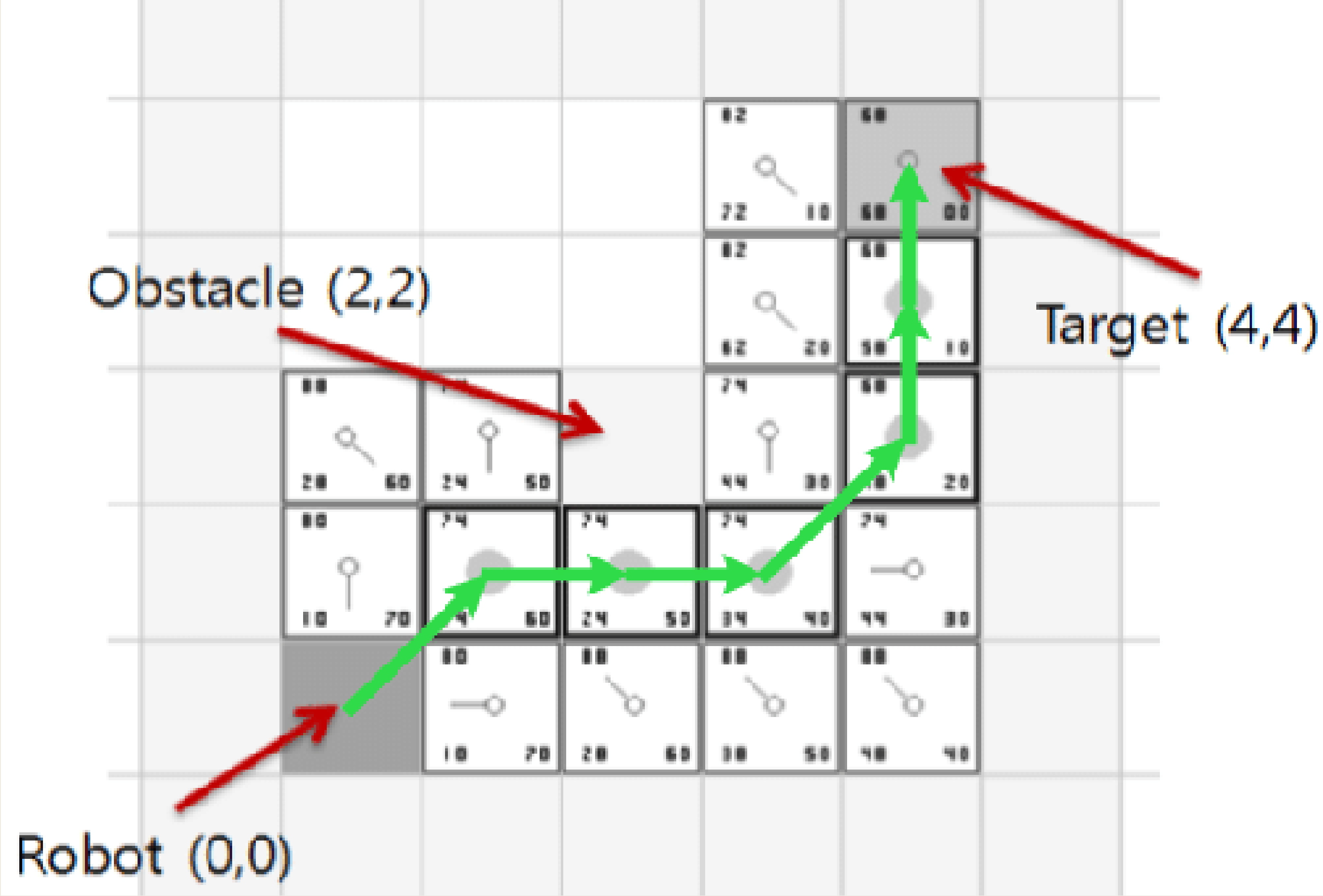
Mapping Techniques

- **Grid Maps:**
  - *Represent the environment as a grid of cells, each indicating free space, obstacles, or unknown areas.*
  - *Indoor robot navigation, path planning.*
- **Topological Maps:**
  - *Represent the environment as a graph of nodes and edges, focusing on connectivity rather than precise geometry.*
  - *High-level navigation, route planning.*
- **Metric Maps:**
  - *Provide precise geometric representations of the environment.*
  - *Detailed navigation, obstacle avoidance.*



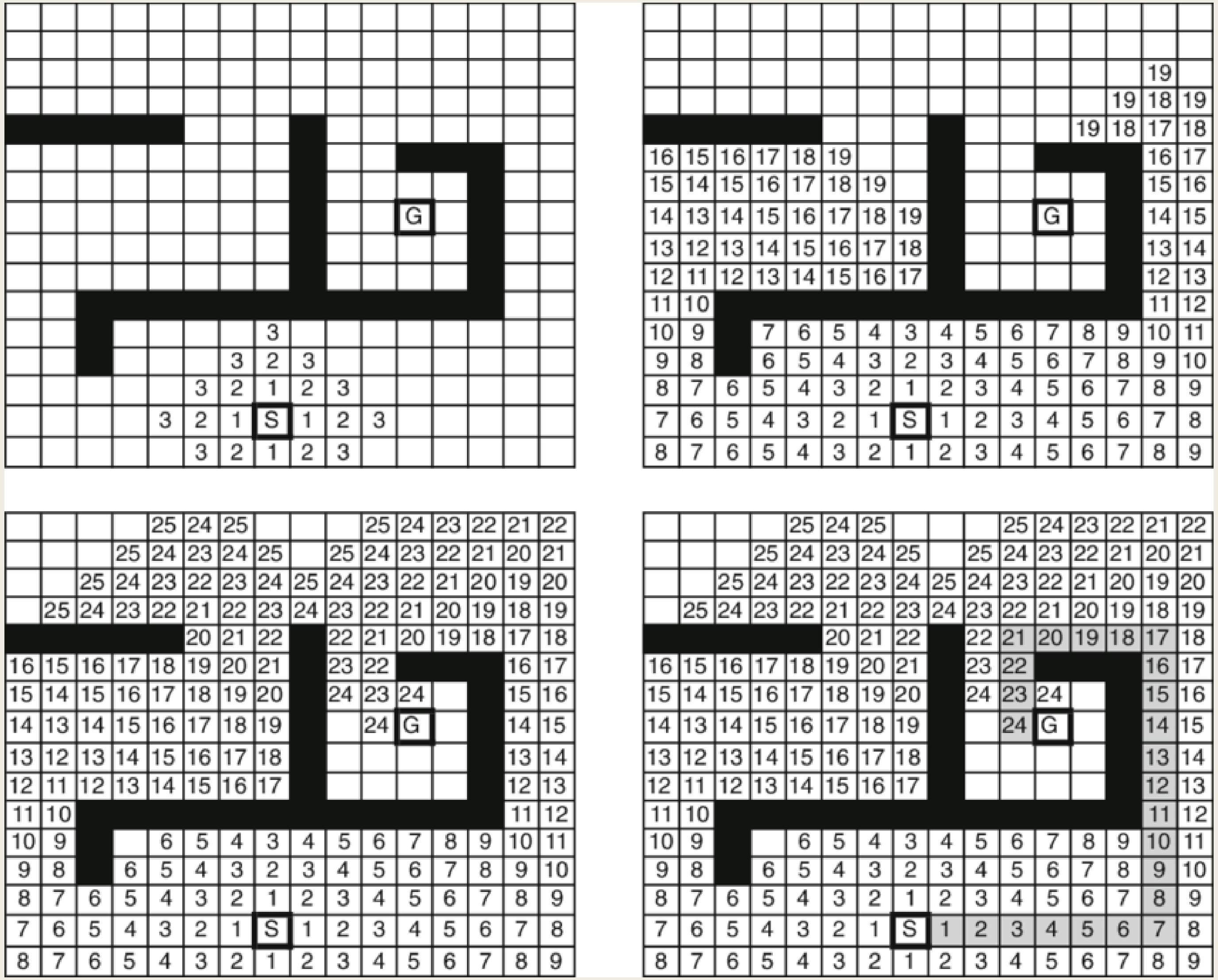
Path Planning Algorithms – A*

- *A* Algorithm:*
  - *Finds the shortest path from the start to the goal using a heuristic to guide the search.*
- Steps:
  - *Initialize open and closed lists.*
  - *Add the start node to the open list.*
  - *Repeat until the goal is reached:*
  - *Select the node with the lowest cost.*
  - *Expand the node, updating costs for neighboring nodes.*
  - *Move the node to the closed list.*



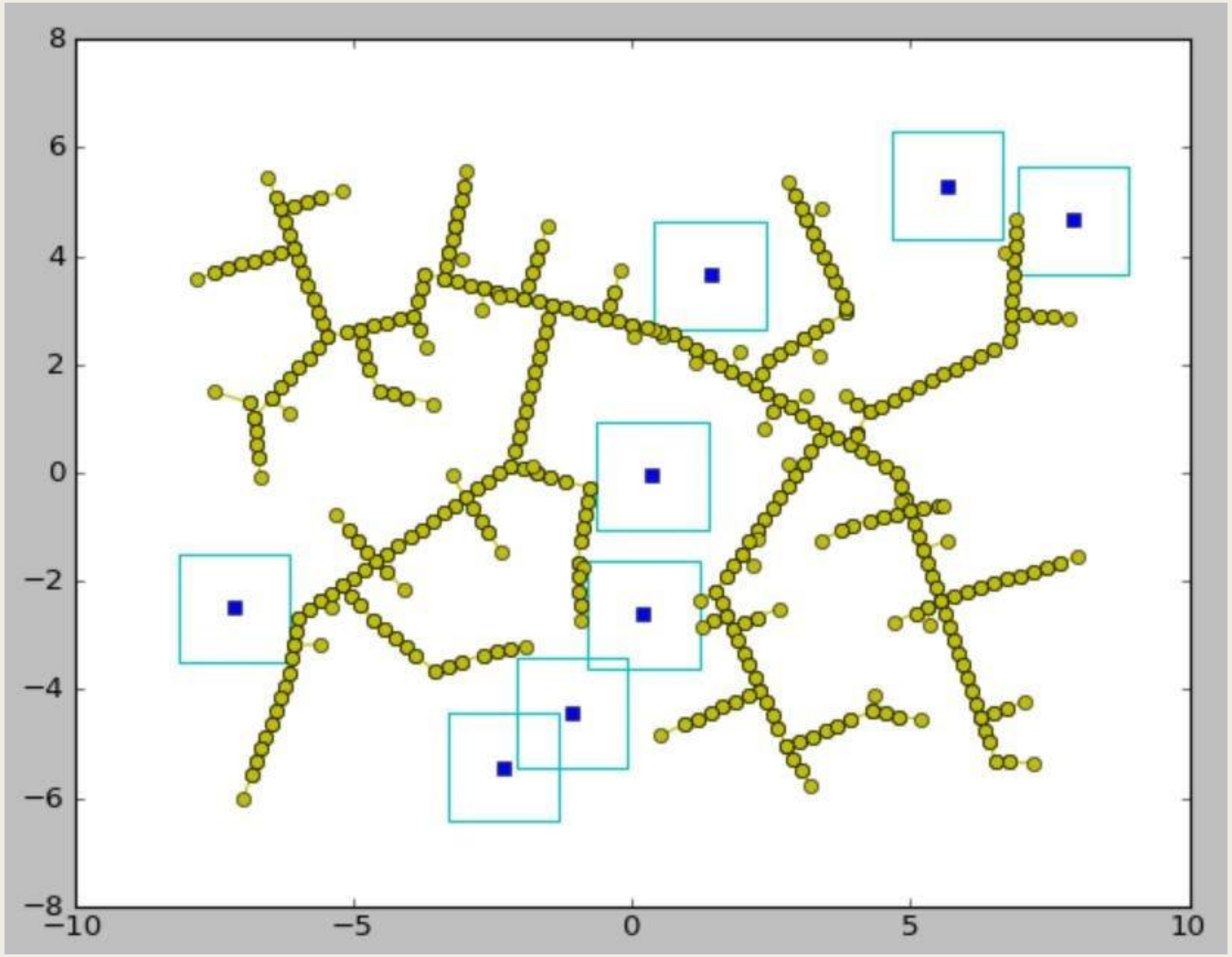
Path Planning Algorithms - Dijkstra

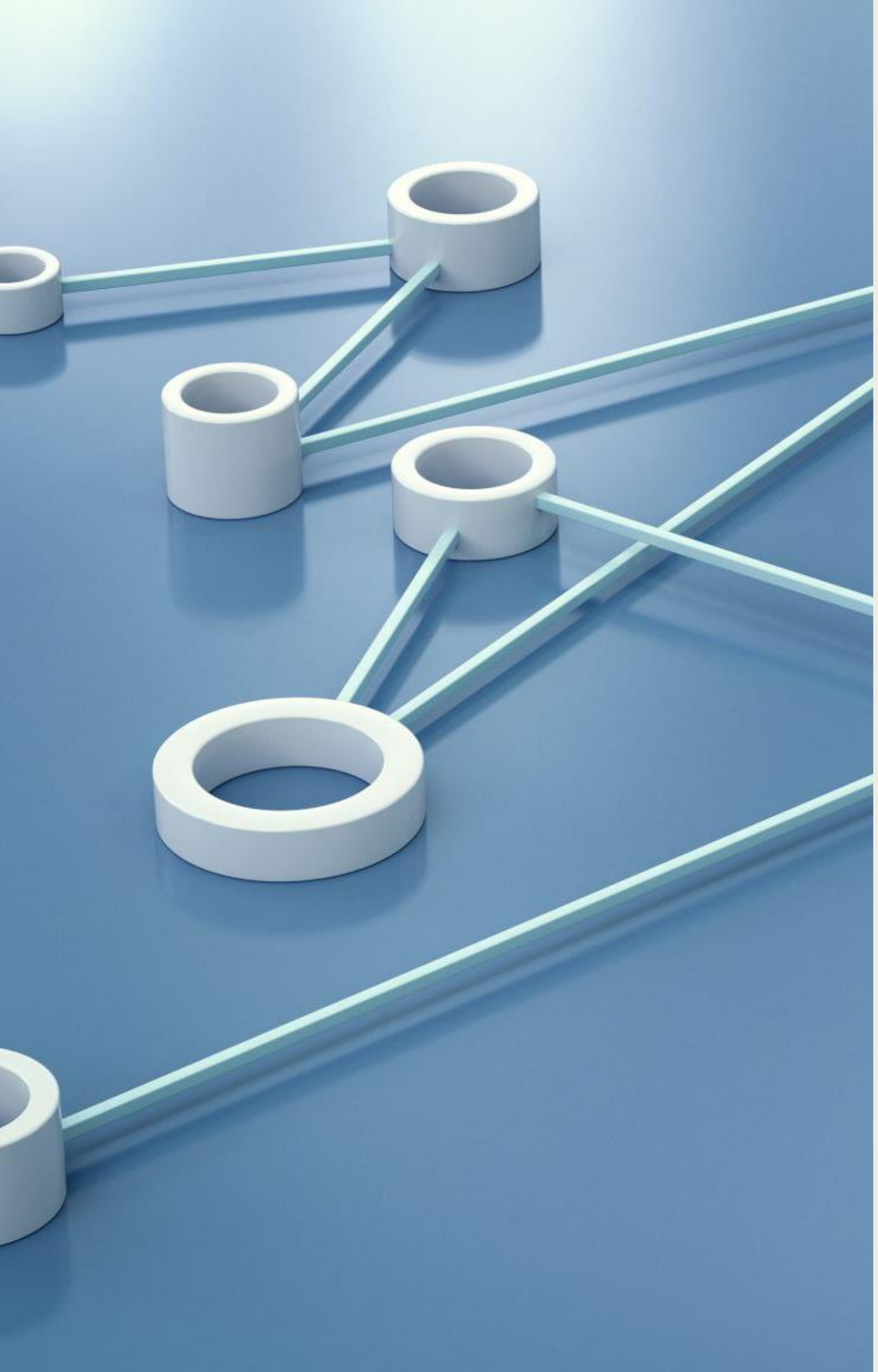
- **Dijkstra's Algorithm:**
  - *Finds the shortest path from the start to all nodes in the graph, ensuring the shortest path to the goal.*
- **Steps:**
  - *Initialize the distances to infinity, except the start node.*
  - *Add the start node to the priority queue.*
  - *Repeat until the queue is empty:*
  - *Extract the node with the smallest distance.*
  - *Update distances for neighboring nodes.*



Path Planning Algorithms - RRT

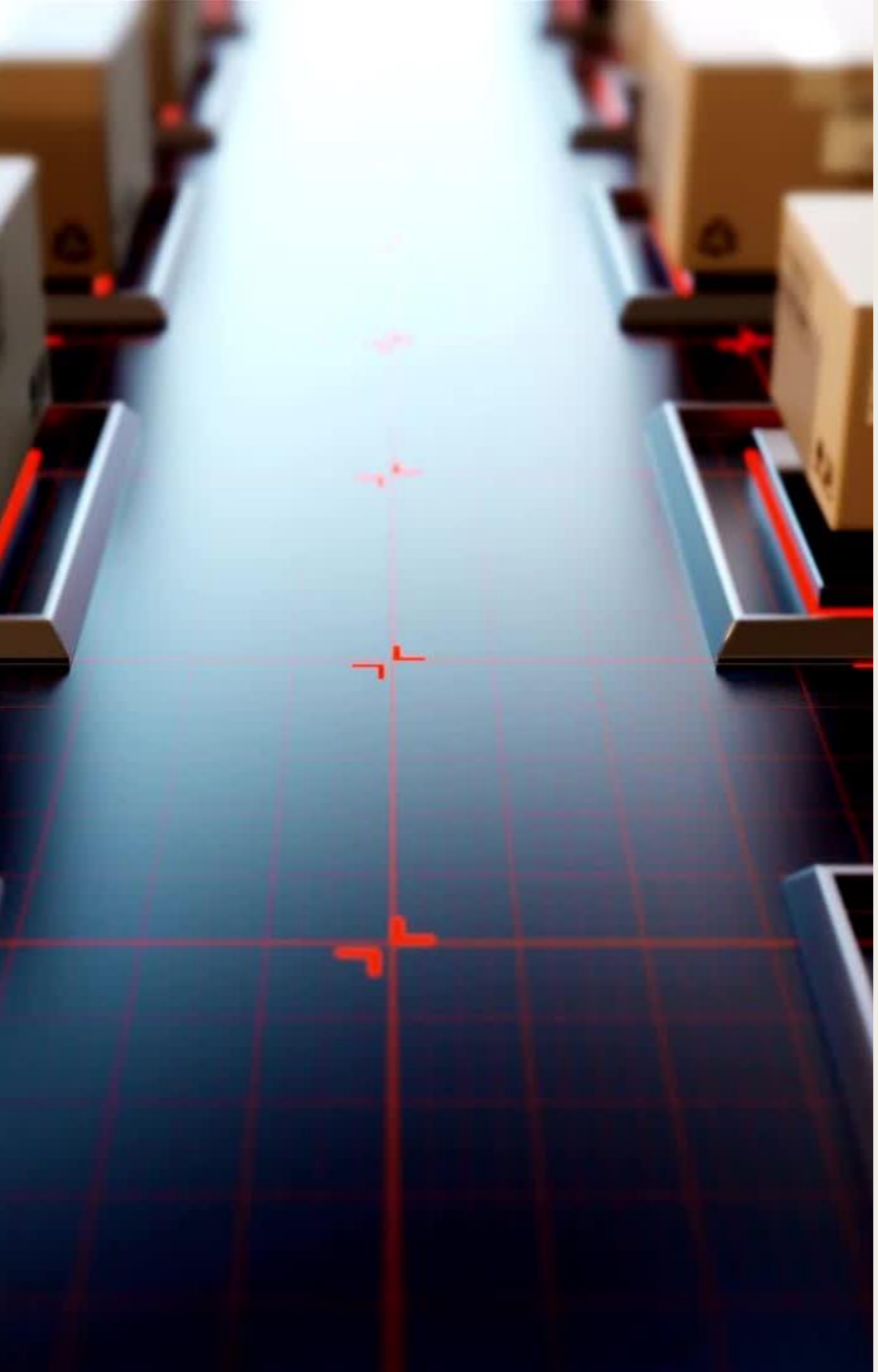
- Builds a tree by randomly sampling points in the space and connecting them to the nearest tree node.
- **Applications:** High-dimensional spaces, dynamic environments.





Challenges in Autonomous Navigation

- **Dynamic Environments:** Navigating and adapting to changing environments in real-time.
- **Sensor Fusion:** Integrating data from multiple sensors for accurate perception and localization.
- **Computational Efficiency:** Ensuring real-time performance with limited computational resources.
- **Safety and Reliability:** Guaranteeing safe and reliable operation in various conditions.
- **Regulatory Compliance:** Adhering to legal and regulatory requirements for autonomous systems.



Summary

- **Components of Autonomous Navigation:** Perception, localization, mapping, path planning, motion control.
- **Techniques and Algorithms:** Sensors, GPS, SLAM, A*, Dijkstra, RRT, PID control, MPC.
- **Applications:** Autonomous vehicles, drones, service robots, industrial automation.
- **Understanding Autonomous Navigation:** Importance of autonomous navigation in enabling robots to operate independently.
- **Challenges and Future Directions:** Current challenges and potential advancements in autonomous navigation.

Q&A